



Iteration/Looping (while loop)

```
Python — vi while.py — 70×19
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Square an integer
#####

x = input('Enter an integer: ')
x = int(x)

ans = 0
itersLeft = x
while (itersLeft != 0):
    ans = ans + x
    itersLeft = itersLeft - 1
print (str(x) + '*' + str(x) + ' = ' + str(ans))
~
~
~
"while.py" 15L, 336C
```

```
Python — -bash — 54×7
Deepaks-MacBook-Pro:Python deepak$ ./while.py
Enter an integer: 7
7*7 = 49
Deepaks-MacBook-Pro:Python deepak$ ./while.py
Enter an integer: 25
25*25 = 625
Deepaks-MacBook-Pro:Python deepak$
```

abs function



```
Python — vi while_mod.py — 70×19
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3
```

```
#####
#Square an integer
#####
```

```
x = input('Enter an integer: ')
x = int(x)
```

```
ans = 0
itersLeft = abs(x)
while (itersLeft != 0):
    ans = ans + abs(x)
    itersLeft = itersLeft - 1
print (str(x) + '*' + str(x) + ' = ' + str(ans))
```

```
~
~
~
```

```
"while_mod.py" 15L, 346C
```

```
Python — -bash — 54×7
```

```
Deepaks-MacBook-Pro:Python deepak$ ./while_mod.py
```

```
Enter an integer: -7
```

```
-7*-7 = 49
```

```
Deepaks-MacBook-Pro:Python deepak$ ./while_mod.py
```

```
Enter an integer: -25
```

```
-25*-25 = 625
```

```
Deepaks-MacBook-Pro:Python deepak$
```

break statement



```
Python — vi break.py — 70x17
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Find an integer divisible by 11 and 12
#####

x = 1

while True:
    if x%11 == 0 and x%12 == 0:
        break
    x = x + 1
print(x, 'is divisible by 11 and 12')
~
~
~
"break.py" 13L, 291C
```

```
Python — -bash — 52x5
Deepaks-MacBook-Pro:Python deepak$ ./break.py
132 is divisible by 11 and 12
Deepaks-MacBook-Pro:Python deepak$
```



pass and continue statements

```
AS2023 — vi pass_continue.py — 70×20
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3
```

```
#####
#continue
#####
for i in range(1,20):
    if i==1:
        pass
        print('This statement is after pass')
    elif i%2==0:
        continue
        print('This statement is after continue')
    elif i==11:
        break
    else:
        print('This is odd')
```

```
~
~
"pass_continue.py"
```

```
AS2023 — -bash — 62×7
```

```
(base) Deepaks-MacBook-Pro:AS2023 deepak$ ./pass_continue.py
```

```
This statement is after pass
```

```
3 is odd
```

```
5 is odd
```

```
7 is odd
```

```
9 is odd
```

```
(base) Deepaks-MacBook-Pro:AS2023 deepak$
```



pass and continue statements

```
AS2023 — vi pass_continue2.py — 70x20
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3
```

```
#####
#continue
#####
for i in range(1,20):
    if i==1:
        pass
        print('This statement is after pass')
    elif i%2==0:
        continue
    elif i==10:
        print('This statement is after pass')
    else:
        print(i, 'is odd')
```

~

~

"pass_continue2.py"

```
AS2023 — -bash — 63x12
```

```
(base) Deepaks-MacBook-Pro:AS2023 deepak$ ./pass_continue2.py
```

```
This statement is after pass
```

```
3 is odd
```

```
5 is odd
```

```
7 is odd
```

```
9 is odd
```

```
11 is odd
```

```
13 is odd
```

```
15 is odd
```

```
17 is odd
```

```
19 is odd
```

```
(base) Deepaks-MacBook-Pro:AS2023 deepak$
```

for loop



```
Python — vi for.py — 87×15
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#for loop
#####

x = 4

for i in range (0, x): #arguments to range function are evaluated before 1st iteration
    print(i)
    x = 5

~
~
~
"for.py" 11L, 268C
```

```
Python — -bash — 52×9
[Deepaks-MacBook-Pro:Python deepak$ ./for.py
0
1
2
3
Deepaks-MacBook-Pro:Python deepak$
```

for loop (take 2)

```
Python — vi for2.py — 72×16
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#for loop (take2)
#####

x = 4

for j in range(x):
    for i in range(x):
        print(i)
        x = 2
~
~
~
"for2.py" 12L, 230C
```

```
Python — -bash — 52×14
Deepaks-MacBook-Pro:Python deepak$ ./for2.py
0
1
2
3
0
1
0
0
1
Deepaks-MacBook-Pro:Python deepak$
```

for loop (take 3)

```
Python — vi for3.py — 74x16
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#for loop (take3)
#####

total = 0

for c in '123456789':      #for statement can be used with in operator
                           to iterate over characters of a string
    total = total + int(c)
print(total)
~
~
~
"for3.py" 11L, 356C
```

```
Python — -bash — 54x7
[Deepaks-MacBook-Pro:Python deepak$ ./for3.py
45
Deepaks-MacBook-Pro:Python deepak$
```


Prime numbers



```
AS2023 — vi prime_number_range_counter.py — 70x22
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#range
#####
lower=int(input("Enter the lower interval: "))
upper=int(input("Enter the upper interval: "))

for num in range(lower,upper+1):
    if num>1:
        j=0
        for i in range(2,num):
            if(num%i)==0:
```

```
AS2023 — -bash — 74x9
(base) Deepaks-MacBook-Pro:AS2023 deepak$ ./prime_number_range_counter.py
Enter the lower interval: 2
Enter the upper interval: 12
3
5
7
11
~
~
"prime_n
(base) Deepaks-MacBook-Pro:AS2023 deepak$
```

for else loop

```
AS2023 — vi for_else.py — 70x14
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#for else
#####

for x in range(6):
    print(x)
else:
    print("Finally finished!")
~
~
~
"for_else.py" 10L,
```

```
AS2023 — -bash — 58x9
(base) Deepaks-MacBook-Pro:AS2023 deepak$ ./for_else.py
0
1
2
3
4
5
Finally finished!
(base) Deepaks-MacBook-Pro:AS2023 deepak$
```

Prime numbers



```
AS2023 — vi prime_number_range_forelse.py — 70x18
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#range
#####
lower=int(input("Enter the lower interval: "))
upper=int(input("Enter the upper interval: "))

for num in range(lower,upper+1):
    if num>1:
        for i in range(2,num):
            if(num%i)==0:
                break
```

```
~
~
"prime_n
AS2023 — -bash — 74x9
(base) Deepaks-MacBook-Pro:AS2023 deepak$ ./prime_number_range_forelse.py
Enter the lower interval: 2
Enter the upper interval: 12
2
3
5
7
11
(base) Deepaks-MacBook-Pro:AS2023 deepak$
```



while else loop

```
AS2023 — vi while_else.py — 70x15
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#while else
#####

i = 1
while i < 6:
    print(i)
    i += 1
else:
    print("i is no longer less than 6")

~
~
"while_else.py"
```

```
AS2023 — -bash — 63x9
(base) Deepaks-MacBook-Pro:AS2023 deepak$ ./while_else.py
1
2
3
4
5
i is no longer less than 6
(base) Deepaks-MacBook-Pro:AS2023 deepak$
```

Practice Programs



PP1: Make a program (without using compound expressions) to determine the grade of the marks entered by the user as per the scheme below:

Marks	Grade
$90 \leq x$	A
$80 \leq x < 90$	B
$70 \leq x < 80$	C
$60 \leq x < 70$	D
$x < 60$	F

PP2: Make a program to find the sum of even and odd digits of a number entered by the user.

Python – Lecture 5
(Functions, File handling)



Deepak Sharma
BSBE and DS&AI
IIT Roorkee



Reserved words



```
Python — vi reserved_words.py — 41×19
Reserved Words
and
or
not
if
else
elif
break
continue
for
in
while
class
with
~
~
~
~
"reserved_words.py" 14L, 78C
```

Cube root (logic 1)

```
Python — vi cube_root.py — 72×22
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Find the cube root of a perfect cube
#####

x = int(input('Enter an integer: '))
ans = 0

while ans**3 < abs(x):
    ans = ans + 1

if ans**3 != abs(x):
    print(x, 'is not a perfect cube')
else:
    if x < 0:
        ans = -ans
    print('Cube root of', x, 'is', ans)

~
~
~
"cube_root.py" 18L, 391C
```

```
Python — -bash — 50×10
Deepaks-MacBook-Pro:Python deepak$ ./cube_root.py
Enter an integer: 27
Cube root of 27 is 3
Deepaks-MacBook-Pro:Python deepak$ ./cube_root.py
Enter an integer: 9
9 is not a perfect cube
Deepaks-MacBook-Pro:Python deepak$ ./cube_root.py
Enter an integer: -8
Cube root of -8 is -2
Deepaks-MacBook-Pro:Python deepak$
```




Cube root (logic 2)

```
Python — vi cube_root2.py — 72x25
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Find the cube root of a perfect cube
#####

x = int(input('Enter an integer: '))

#ans = 0
#while ans**3 < abs(x):
#    ans = ans + 1

for ans in range(0, abs(x)+1):
    if ans**3 >= abs(x):
        break

if ans**3 != abs(x):
    print(x, 'is not a perfect cube')
else:
    if x < 0:
        ans = -ans
    print('Cube root of', x, 'is', ans)

~
~
"cube_root2.py" 22L, 456C
```

```
Python — -bash — 50x10
Deepaks-MacBook-Pro:Python deepak$ ./cube_root.py
Enter an integer: 27
Cube root of 27 is 3
Deepaks-MacBook-Pro:Python deepak$ ./cube_root.py
Enter an integer: 9
9 is not a perfect cube
Deepaks-MacBook-Pro:Python deepak$ ./cube_root.py
Enter an integer: -8
Cube root of -8 is -2
Deepaks-MacBook-Pro:Python deepak$
```